

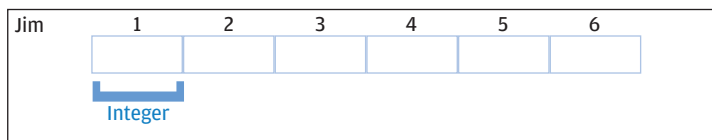
Compound Data Types

8.1 Arrays

Arrays are defined as a series of elements of the same type, placed in contiguous memory locations. They are referenced individually, by adding an index to a unique identifier. An array can also be defined as a data structure, allowing a collective name to be given to a group of elements having the same type. The individual element of an array is identified by its own unique index. This is also referred as the subscript of the element.

If we want to store six values of the type integer, but don't want to declare a variable with a different identifier for each of them, we can do it by using arrays.

This is shown as follows:



Here, the array `Jim` is of type `integer` that contains six integer values and is represented as shown above. Each blank panel shown above represents an individual element of the array `Jim`. The elements of the array are numbered from 0 to 5. These numbers are called index of the array element. Note that in arrays, the first index is always 0. An array is like a regular variable and should be declared before it can be used. A normal array declaration is as follows: